

Inverse Subfactorials at Infinity

Lambert- W carriers, Bell-number sectors, and a general calculus of gamma-dominant transseries

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Abstract

The subfactorial sequence $D_n = !n$, which counts derangements, is discrete and therefore has no unique inverse until an interpolation or a generalized-inverse convention is chosen. This article resolves that ambiguity explicitly and derives asymptotic inverse transseries for several canonical choices. For every branch index $\kappa \in \mathbb{Z}$, the incomplete-gamma continuation admits the exact splitting

$$\mathfrak{D}_\kappa(z) = \frac{\Gamma(z+1)}{e} + e^{(2\kappa+1)\pi iz} C(z), \quad C(z) = \frac{1}{e} \int_0^1 t^z e^t dt.$$

The correction has the Bell-number expansion

$$C(x) \sim \sum_{m \geq 1} (-1)^{m-1} \frac{\mathcal{B}_m}{x^m},$$

with an exact signed remainder bounded by $\mathcal{B}_{M+1}x^{-M-1}$ after M terms. Let $u(Y)$ be the large real solution of $\Gamma(u+1)/e = Y$, put

$$A = \log \frac{eY}{\sqrt{2\pi}}, \quad w = W_0(A/e), \quad Q = \frac{A}{w}, \quad s = 1 + w.$$

We derive the all-orders gamma-envelope inverse

$$u(Y) \sim Q - \frac{1}{2} + \frac{1}{24sQ} - \frac{14s^2 + 10s + 5}{5760s^3Q^3} + \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5Q^5} + \dots$$

The inverse of the full branch is then a nested transseries

$$\mathfrak{D}_\kappa^{-1}(Y) \sim u(Y) + \sum_{r \geq 1} \frac{e^{r(2\kappa+1)\pi i u(Y)}}{Y^r} \Phi_r^{(\kappa)}(u(Y)),$$

where every coefficient is given by an explicit finite differential-operator product. The first coefficient is $-C/\psi$, and hence the first genuinely subfactorial displacement has size $Y^{-1}/(u \log u)$, far below every fixed exponential scale e^{-cu} . We also obtain the inverse of the canonical real symmetrization $\Gamma(x+1)/e + C(x) \cos(\pi x)$, parity-resolved inverses, and the discrete staircase inverse. Finally, we develop a general two-stage inversion theory for dominant logarithmic-Lambert carriers perturbed by algebraic, oscillatory, or multi-frequency moment sectors, including a Touchard-polynomial deformation of the Bell expansion.

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1 Introduction

1.1 Why the phrase “inverse subfactorial” needs a definition

The derangement number

$$D_n = !n = n! \sum_{j=0}^n \frac{(-1)^j}{j!}$$

is defined on the nonnegative integers. A sequence does not possess an ordinary inverse function on a continuum. Several natural inverse objects coexist:

1. the generalized inverse of the increasing integer sequence, which is a staircase;
2. a real smooth interpolation agreeing with every D_n ;
3. the complex incomplete-gamma continuation $\Gamma(z + 1, -1)/e$, after a branch is selected;
4. the two nonoscillatory parity envelopes that interpolate the even and odd subsequences separately.

These objects have different asymptotic structures. The staircase inverse has an unavoidable order-one rounding sector. The smooth and analytic inverses have much finer transseries whose first subfactorial correction is superexponentially small relative to the inverse variable.

The main purpose of this article is to derive all of these structures without conflating them. The formal-calculus viewpoint is compatible with the polynomial-logarithmic reversion framework collected in the ProveIt series-and-transseries directory [10]. The special feature here is that the derangement correction is not merely another inverse power: relative to $n!/e$, it is a new, enormously smaller sector whose coefficients are Bell numbers.

1.2 The four scales

Write $Y \rightarrow +\infty$ for the target value, and let $u = u(Y)$ solve

$$\frac{\Gamma(u + 1)}{e} = Y.$$

Then $u \asymp \log Y / \log \log Y$. Four asymptotic layers should be distinguished.

1. **Lambert-logarithmic layer.** The leading inverse is expressed by a Lambert W carrier and may be expanded in nested logarithms.
2. **Stirling inverse-power layer.** Centered Stirling corrections produce powers $u^{-1}, u^{-3}, u^{-5}, \dots$, divided by powers of $1 + W$.
3. **Hyperasymptotic gamma layer.** The divergent Stirling series has a least-term scale of order $u^{-1/2}e^{-2\pi u}$. A complete numerical resolution below that scale requires an exact or hyperasymptotic treatment of the gamma envelope.
4. **Bell/subfactorial layer.** The additive derangement correction changes the inverse by

$$O\left(\frac{1}{Y u \log u}\right) = \exp\{-u \log u + O(u)\},$$

which is smaller than e^{-cu} for every fixed $c > 0$.

This hierarchy explains why the cleanest final answer is a *nested* transseries: first keep the inverse-gamma carrier $u(Y)$ exact, then expand the Bell sector in powers of Y^{-1} . The Lambert–Stirling expansion of u can be inserted afterward to any desired algebraic accuracy.

1.3 Notation and conventions

Bell numbers are denoted by $\mathcal{B}_m$, so that $\mathcal{B}_0 = 1$, $\mathcal{B}_1 = 1$, $\mathcal{B}_2 = 2$, $\mathcal{B}_3 = 5$, and so on. Bernoulli numbers and polynomials retain the conventional notation B_m and $B_m(x)$. The digamma and polygamma functions are

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}, \quad \psi^{(j)}(z) = \frac{d^j}{dz^j} \psi(z).$$

The principal Lambert branch is W_0 . Unless a complex sector is explicitly discussed, all logarithms and all large variables are real and positive.

For a formal series in t , $[t^n]H(t)$ denotes coefficient extraction. Asymptotic statements are Poincaré statements: the number of retained terms is fixed before the large-variable limit is taken.

2 Derangements, incomplete gamma, and the Bell correction

2.1 Classical identities

A derangement is a permutation without fixed points. Inclusion-exclusion gives

$$D_n = n! \sum_{j=0}^n \frac{(-1)^j}{j!}. \tag{2.1}$$

The upper incomplete gamma identity

$$\Gamma(n+1, z) = n! e^{-z} \sum_{j=0}^n \frac{z^j}{j!}$$

therefore yields

$$D_n = \frac{\Gamma(n+1, -1)}{e}. \quad (2.2)$$

The right-hand side requires a branch convention away from integer n . That branch information is precisely what creates the oscillatory factor in the inverse transseries.

Hassani proved the exact representation

$$D_n = \frac{n!}{e} + (-1)^n \frac{L_n(e)}{e}, \quad L_n(e) = \int_1^e (\log t)^n dt, \quad (2.3)$$

and derived the Bell-number asymptotic expansion with an explicit remainder bound [1]. We now place that result inside an analytic continuation that is adapted to inversion.

2.2 A branch-indexed exact decomposition

For $\kappa \in \mathbb{Z}$, set

$$\theta_\kappa = (2\kappa + 1)\pi, \quad \omega_\kappa = i\theta_\kappa.$$

When the segment from 0 to -1 is traversed with argument θ_κ , define the corresponding branch of the lower incomplete gamma by that path. The standard relation $\Gamma(s, -1) = \Gamma(s) - \gamma(s, -1)$ then gives the following exact formula.

Theorem 2.1 (Exact branch splitting). *For $\Re z > -1$, define*

$$F(z) = \frac{\Gamma(z+1)}{e}, \quad C(z) = \frac{1}{e} \int_0^1 t^z e^t dt = \frac{1}{e} \sum_{j=0}^{\infty} \frac{1}{j!(z+j+1)}. \quad (2.4)$$

Then the branch-indexed subfactorial continuation is

$$\boxed{\mathfrak{D}_\kappa(z) = F(z) + e^{\omega_\kappa z} C(z)}. \quad (2.5)$$

For every nonnegative integer n , this value is independent of κ and equals D_n .

Proof. Put $s = z + 1$ and parametrize the chosen path by $t = re^{i\theta_\kappa}$, $0 \leq r \leq 1$. Since $e^{i\theta_\kappa} = -1$,

$$\gamma(s, -1) = e^{i\theta_\kappa s} \int_0^1 r^{s-1} e^r dr.$$

Consequently,

$$\begin{aligned} \frac{\Gamma(z+1, -1)}{e} &= \frac{\Gamma(z+1)}{e} - \frac{e^{i\theta_\kappa(z+1)}}{e} \int_0^1 r^z e^r dr \\ &= F(z) + e^{i\theta_\kappa z} C(z), \end{aligned}$$

because $e^{i\theta_\kappa} = -1$. Expanding e^t under the integral proves the convergent partial-fraction series in Equation (2.4). At an integer n , the phase is $(-1)^n$. After the substitution $t = e^v$, one has

$$L_n(e) = \int_0^1 v^n e^v dv = eC(n),$$

so Equation (2.3) completes the identification with D_n . $\square$

Remark 2.2. Formula Equation (2.5) may be taken as the definition in the half-plane $\Re z > -1$, avoiding any ambiguity in incomplete-gamma path notation. The cancellation that continues the function farther left is not needed for the large-argument analysis.

2.3 The canonical real symmetrization

The two adjacent branches $\kappa = 0$ and $\kappa = -1$ are complex conjugates on the real axis. Their average is therefore a distinguished real interpolation.

Definition 2.3 (Symmetrized real subfactorial). For real $x > -1$, set

$$\mathfrak{D}_{\text{sym}}(x) = \frac{\mathfrak{D}_0(x) + \mathfrak{D}_{-1}(x)}{2} = F(x) + C(x) \cos(\pi x). \quad (2.6)$$

Then $\mathfrak{D}_{\text{sym}}(n) = D_n$ for every nonnegative integer n .

The following elementary fact justifies the real inverse and, later, the ceiling formula for the discrete inverse.

Proposition 2.4 (Eventual monotonicity of the real interpolation). *There exists $X > 0$ such that $\mathfrak{D}'_{\text{sym}}(x) > 0$ for every $x \geq X$. Consequently, $\mathfrak{D}_{\text{sym}}$ has a unique terminal real inverse.*

Proof. From the integral representation, $C(x) = O(x^{-1})$ and $C'(x) = O(x^{-2})$. Therefore

$$\frac{d}{dx}(C(x) \cos(\pi x)) = C'(x) \cos(\pi x) - \pi C(x) \sin(\pi x) = O(x^{-1}).$$

On the other hand,

$$F'(x) = F(x)\psi(x+1) \sim F(x) \log x \longrightarrow +\infty.$$

The positive gamma-envelope derivative thus eventually dominates the oscillatory derivative. $\square$

It is also useful to introduce the parity envelopes

$$G_{\sigma}(x) = F(x) + \sigma C(x), \quad \sigma \in \{+1, -1\}. \quad (2.7)$$

For even n , $G_+(n) = D_n$; for odd n , $G_-(n) = D_n$. These envelopes remove the oscillatory phase and isolate the two lattice subsequences.

2.4 Bell numbers from a Stieltjes kernel

The correction C is a Stieltjes transform of a positive discrete measure. Let

$$d\mu(t) = \frac{1}{e} \sum_{j=0}^{\infty} \frac{1}{j!} \delta_{j+1}(dt). \quad (2.8)$$

Then

$$C(x) = \int_0^{\infty} \frac{d\mu(t)}{x+t}.$$

The moments of this measure are Bell numbers shifted by one index:

$$\int_0^{\infty} t^{m-1} d\mu(t) = \frac{1}{e} \sum_{j=0}^{\infty} \frac{(j+1)^{m-1}}{j!} = \mathcal{B}_m, \quad m \geq 1. \quad (2.9)$$

The last equality is Dobinski's formula after shifting the summation index.

Theorem 2.5 (Exact Bell expansion and signed remainder). *For every real $x > 0$ and integer $M \geq 1$,*

$$C(x) = \sum_{m=1}^M (-1)^{m-1} \frac{\mathcal{B}_m}{x^m} + R_M(x), \quad (2.10)$$

where

$$R_M(x) = \frac{(-1)^M}{e x^M} \sum_{j=0}^{\infty} \frac{(j+1)^M}{j!(x+j+1)}. \quad (2.11)$$

In particular,

$$(-1)^M R_M(x) > 0, \quad |R_M(x)| \leq \frac{\mathcal{B}_{M+1}}{x^{M+1}}. \quad (2.12)$$

Thus

$$C(x) \sim \sum_{m \geq 1} (-1)^{m-1} \frac{\mathcal{B}_m}{x^m}. \quad (2.13)$$

Proof. For $b > 0$, finite geometric division gives the exact identity

$$\frac{1}{x+b} = \sum_{m=1}^M (-1)^{m-1} \frac{b^{m-1}}{x^m} + (-1)^M \frac{b^M}{x^M(x+b)}. \quad (2.14)$$

Set $b = j+1$, multiply by $1/(ej!)$, and sum over $j \geq 0$. The moment identity Equation (2.9) gives the finite sum in Equation (2.10), and the last term gives Equation (2.11). Positivity proves the sign. Since $x+j+1 \geq x$,

$$|R_M(x)| \leq \frac{1}{e x^{M+1}} \sum_{j=0}^{\infty} \frac{(j+1)^M}{j!} = \frac{\mathcal{B}_{M+1}}{x^{M+1}}.$$

□

The first terms are

$$C(x) \sim \frac{1}{x} - \frac{2}{x^2} + \frac{5}{x^3} - \frac{15}{x^4} + \frac{52}{x^5} - \frac{203}{x^6} + \cdots. \quad (2.15)$$

The series diverges for every fixed x : for example, $\mathcal{B}_{2m} \geq (2m-1)!!$, so $\mathcal{B}_m^{1/m}$ is unbounded. Its role is asymptotic, and the explicit remainder bound tells us exactly how a fixed truncation behaves as $x \rightarrow \infty$.

2.5 Derivatives and differential closure

Differentiation under the integral gives

$$C^{(r)}(x) = \frac{1}{e} \int_0^1 t^x (\log t)^r e^t dt, \quad (-1)^r C^{(r)}(x) > 0. \quad (2.16)$$

Termwise differentiation of the Poincare expansion is legitimate at every fixed order and yields

$$C^{(r)}(x) \sim \sum_{m \geq 1} (-1)^{m+r-1} (m)_r \mathcal{B}_m x^{-m-r}, \quad (2.17)$$

where $(m)_r = m(m+1) \cdots (m+r-1)$ is the rising factorial. These formulas will place all inverse-sector coefficients in a differential ring generated by inverse powers, inverse logarithms, Bell numbers, and Bernoulli numbers.

3 Which inverse is being expanded?

3.1 The dominant gamma-envelope inverse

Let

$$F(x) = \frac{\Gamma(x+1)}{e}. \quad (3.1)$$

Because $\psi(x+1) \sim \log x$, there is an x_0 such that $F'(x) > 0$ for $x \geq x_0$. We denote the inverse on this terminal interval by

$$u(Y) = F^{-1}(Y), \quad Y \geq F(x_0). \quad (3.2)$$

This exact function will be the carrier for every finer inverse. At $x = u(Y)$, introduce

$$\lambda = \lambda(u) = \psi(u+1), \quad P_k = P_k(u) = \frac{F^{(k)}(u)}{F(u)}. \quad (3.3)$$

The first normalized derivatives are

$$P_1 = \lambda, \quad P_2 = \lambda^2 + \lambda', \quad P_3 = \lambda^3 + 3\lambda\lambda' + \lambda'', \quad (3.4)$$

and in general

$$P_0 = 1, \quad P_{k+1} = P'_k + \lambda P_k. \quad (3.5)$$

Thus P_k is the complete exponential Bell polynomial in $\lambda, \lambda', \dots$. Since $F(u) = Y$,

$$F^{(k)}(u) = Y P_k(u), \quad F'(u) = Y \lambda(u). \quad (3.6)$$

3.2 Large inverse branches of the analytic continuation

Fix κ . For large positive Y , Equation (2.5) has a unique solution $z_\kappa(Y)$ in a sufficiently small disk around $u(Y)$. Indeed,

$$\mathfrak{D}_\kappa(u) - Y = e^{\omega_\kappa u} C(u) = O(u^{-1}),$$

whereas

$$\mathfrak{D}'_\kappa(u) = Y \lambda + O(u^{-1}).$$

Consequently,

$$z_\kappa(Y) - u(Y) = O\left(\frac{1}{Y u \log u}\right). \quad (3.7)$$

The root is generally complex even when Y is real. Different incomplete-gamma path branches give different phases away from integer arguments, although all of them agree on the derangement lattice.

3.3 The large real inverse

Let

$$r(Y) = \mathfrak{D}_{\text{sym}}^{-1}(Y) \quad (3.8)$$

be the eventual real inverse of Equation (2.6). It is the most natural single-valued real interpolation inverse because it agrees with the complete derangement sequence, not only one parity. Its correction

is oscillatory through $\cos(\pi u)$ and generates higher Fourier harmonics at successive transseries orders.

For parity-resolved questions, let

$$x_\sigma(Y) = G_\sigma^{-1}(Y), \quad \sigma \in \{+1, -1\}. \quad (3.9)$$

Then

$$x_{(-1)^n}(D_n) = n \quad (3.10)$$

for every sufficiently large integer n . The parity inverses are especially convenient for displaying the Bell sector because no trigonometric phase remains.

3.4 The discrete generalized inverse

Since D_n is strictly increasing from $n = 2$ onward, one may define

$$N(Y) = \min \{n \geq 2 : D_n \geq Y\}. \quad (3.11)$$

For all sufficiently large Y , monotonicity of $\mathfrak{D}_{\text{sym}}$ and the interpolation identity give

$$\boxed{N(Y) = \lceil r(Y) \rceil}. \quad (3.12)$$

Indeed, once n is large enough for [Theorem 2.4](#) to apply, the map $\mathfrak{D}_{\text{sym}}$ carries $[n-1, n]$ bijectively onto $[D_{n-1}, D_n]$. Thus $D_{n-1} < Y \leq D_n$ is equivalent to $n-1 < r(Y) \leq n$, with the endpoint convention agreeing with the ceiling.

This inverse contains an order-one discontinuous sector. Away from integer values of $r(Y)$, the classical Fourier representation of the sawtooth gives

$$N(Y) = r(Y) + \frac{1}{2} + \frac{1}{\pi} \sum_{m=1}^{\infty} \frac{\sin(2\pi m r(Y))}{m}. \quad (3.13)$$

At a jump, the Fourier series takes the midpoint value and the ceiling convention must be imposed separately. Formula [Equation \(3.13\)](#) explains why no continuous asymptotic expansion can approximate the staircase uniformly with an error $o(1)$: lattice inversion and smooth transseries inversion are distinct problems.

4 The Lambert–W carrier for the gamma envelope

4.1 Centered Stirling expansion

The shifted form of Stirling’s expansion is most efficient here. Put

$$q = x + \frac{1}{2}. \quad (4.1)$$

The DLMF shifted expansion [\[2\]](#) with shift $1/2$ gives

$$\log \Gamma\left(q + \frac{1}{2}\right) \sim q \log q - q + \frac{1}{2} \log(2\pi) + R(q), \quad (4.2)$$

where

$$R(q) \sim \sum_{m \geq 1} \frac{B_{2m}(1/2)}{2m(2m-1)q^{2m-1}}. \quad (4.3)$$

The identity $B_{2m}(1/2) = (2^{1-2m} - 1)B_{2m}$ shows that only odd inverse powers occur. Explicitly,

$$R(q) \sim -\frac{1}{24q} + \frac{7}{2880q^3} - \frac{31}{40320q^5} + \frac{127}{215040q^7} - \frac{511}{608256q^9} + \cdots. \quad (4.4)$$

For the equation $F(x) = Y$, define

$$A = A(Y) = \log \frac{eY}{\sqrt{2\pi}}. \quad (4.5)$$

Then $q = u + 1/2$ satisfies the exact equation

$$A = q(\log q - 1) + \rho(q), \quad (4.6)$$

where $\rho(q)$ is the exact centered Stirling remainder function and has the asymptotic expansion $R(q)$ in Equation (4.3).

4.2 Exact solution of the leading equation

Discarding ρ gives

$$A = Q(\log Q - 1). \quad (4.7)$$

This equation is solved exactly by Lambert W . Set

$$w = W_0(A/e), \quad Q = \frac{A}{w} = e^{w+1}, \quad s = 1 + w = \log Q. \quad (4.8)$$

Indeed, $we^w = A/e$, so $Qw = A$ and $\log Q - 1 = w$. The familiar leading approximation to the inverse gamma function is therefore

$$u(Y) \sim Q - \frac{1}{2}. \quad (4.9)$$

The shift by $-1/2$ is not cosmetic: it removes the logarithmic prefactor in Stirling's formula and forces the subsequent algebraic correction to begin at order Q^{-1} .

4.3 A general triangular reversion theorem

The following formal result will be reused for other gamma-dominant functions.

Theorem 4.1 (Lambert-centered reversion). *Let $S \rightarrow +\infty$, let a be fixed, and suppose q is determined by*

$$S = q(\log q - a) + R(q), \quad R(q) \sim \sum_{m \geq 1} r_m q^{-m}. \quad (4.10)$$

Define

$$w = W_0(Se^{-a}), \quad Q = \frac{S}{w} = e^{a+w}, \quad s = 1 + w. \quad (4.11)$$

Then there is a unique formal expansion

$$q \sim Q + \sum_{n \geq 1} d_n(s) Q^{-n}. \quad (4.12)$$

The coefficients are determined triangularly. Put $t = Q^{-1}$ and

$$U(t) = \frac{q - Q}{Q} = \sum_{n \geq 1} d_n t^{n+1}. \quad (4.13)$$

Then U is the unique formal solution of

$$0 = t^{-1} \left[sU + \sum_{k \geq 2} \frac{(-1)^k}{k(k-1)} U^k \right] + \sum_{m \geq 1} r_m t^m (1 + U)^{-m}. \quad (4.14)$$

If $U_{n-1} = \sum_{j=1}^{n-1} d_j t^{j+1}$, then

$$d_n = -\frac{1}{s} [t^n] \left\{ t^{-1} \left[sU_{n-1} + \sum_{k \geq 2} \frac{(-1)^k}{k(k-1)} U_{n-1}^k \right] + \sum_{m \geq 1} r_m t^m (1 + U_{n-1})^{-m} \right\}. \quad (4.15)$$

For every fixed N , truncation after $d_N Q^{-N}$ has error

$$O\left(\frac{Q^{-N-1}}{s}\right), \quad (4.16)$$

provided the expansion of R is valid to the corresponding order.

Proof. The carrier identity is $S = Q(\log Q - a) = Qw$. Write $q = Q(1 + U)$. Then

$$\begin{aligned} q(\log q - a) - Q(\log Q - a) &= Q[(1 + U)(w + \log(1 + U)) - w] \\ &= Q[wU + (1 + U)\log(1 + U)]. \end{aligned}$$

The Taylor identity

$$(1 + U)\log(1 + U) = U + \sum_{k \geq 2} \frac{(-1)^k}{k(k-1)} U^k \quad (4.17)$$

turns the last expression into the first term of Equation (4.14); substituting $q^{-m} = t^m(1 + U)^{-m}$ gives the rest. The coefficient of d_n in $[t^n]$ is exactly s , while every nonlinear occurrence involves only $d_1, \dots, d_{n-1}$. Since $s \neq 0$ for large positive S , the recursion is unique.

For the analytic error, let q_N be the truncation and let $\mathcal{R}_N$ be the residual in Equation (4.10). Coefficient matching gives $\mathcal{R}_N = O(Q^{-N-1})$. The derivative of the left-hand side with respect to q is

$$\log q - a + 1 + R'(q) = s + o(s).$$

The mean-value theorem therefore transfers residual order into the error estimate Equation (4.16). $\square$

The first universal coefficients are

$$d_1 = -\frac{r_1}{s}, \quad (4.18)$$

$$d_2 = -\frac{r_2}{s}, \quad (4.19)$$

$$d_3 = -\frac{r_3}{s} - \frac{r_1^2}{s^2} - \frac{r_1^2}{2s^3}, \quad (4.20)$$

$$d_4 = -\frac{r_4}{s} - \frac{3r_1r_2}{s^2} - \frac{r_1r_2}{s^3}. \quad (4.21)$$

The appearance of $s^{-1} = (1 + W_0)^{-1}$ is the formal signature of reversion: it is the reciprocal derivative of the Lambert carrier.

4.4 Explicit inverse-gamma coefficients

For the centered gamma expansion, $r_{2m} = 0$. Equation (4.14) is then odd in the formal grading, so $d_{2m} = 0$. Substituting Equation (4.4) into the recursion gives

$$d_1 = \frac{1}{24s}, \quad (4.22)$$

$$d_3 = -\frac{14s^2 + 10s + 5}{5760s^3}, \quad (4.23)$$

$$d_5 = \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5}, \quad (4.24)$$

$$d_7 = -\frac{822960s^6 + 292536s^5 + 136956s^4 + 68040s^3 + 32970s^2 + 12250s + 2625}{1393459200s^7}. \quad (4.25)$$

Principal gamma-envelope inverse

With A, w, Q, s defined by Equations (4.5) and (4.8),

$$\begin{aligned} u(Y) = Q - \frac{1}{2} + \frac{1}{24sQ} - \frac{14s^2 + 10s + 5}{5760s^3Q^3} \\ + \frac{2232s^4 + 1176s^3 + 714s^2 + 350s + 105}{2903040s^5Q^5} + O\left(\frac{1}{sQ^7}\right). \end{aligned} \quad (4.26)$$

The Q^{-7} term is obtained by inserting Equation (4.25).

The expansion is algorithmic to arbitrary order. Every coefficient is a rational function of $s = 1 + W_0(A/e)$ with rational coefficients because the centered Stirling coefficients are rational.

4.5 Flattening the Lambert carrier into nested logarithms

Set

$$L_1 = \log(A/e) = \log A - 1, \quad L_2 = \log L_1. \quad (4.27)$$

The large-argument expansion of Lambert W , as developed by Corless, Gonnet, Hare, Jeffrey, and Knuth [4], implies

$$\frac{1}{W_0(A/e)} \sim \frac{1}{L_1} + \frac{L_2}{L_1^2} + \frac{L_2^2 - L_2}{L_1^3}$$

$$+ \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^4} + \frac{L_2^4 - \frac{13}{3}L_2^3 + \frac{9}{2}L_2^2 - L_2}{L_1^5} + \dots \quad (4.28)$$

Hence

$$Q \sim A \left[\frac{1}{L_1} + \frac{L_2}{L_1^2} + \frac{L_2^2 - L_2}{L_1^3} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^4} + \dots \right]. \quad (4.29)$$

Combining Equation (4.29) with Equation (4.26) produces a fully flattened polynomial-logarithmic transseries in A , $\log A$, and $\log \log A$. It is usually preferable to keep the exact W_0 carrier, because doing so packages infinitely many nested-log terms and sharply improves numerical conditioning.

4.6 The least-term barrier and why nesting matters

The Bernoulli-number identity

$$B_{2m} = (-1)^{m-1} \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m)$$

shows that the centered Stirling series is factorially divergent. Its least term occurs near $m \sim \pi Q$ and has magnitude

$$O\left(Q^{-1/2} e^{-2\pi Q}\right). \quad (4.30)$$

By contrast, the first inverse-subfactorial sector derived below has size

$$\frac{1}{YQ \log Q} = \exp\{-Q(\log Q - 1) + O(\log Q)\}. \quad (4.31)$$

Therefore

$$\frac{1}{YQ \log Q} = o(e^{-cQ}) \quad \text{for every fixed } c > 0. \quad (4.32)$$

A finite or even optimally truncated Stirling expansion of $u(Y)$ cannot, by itself, numerically resolve the Bell displacement for very large Y . One should first compute $u(Y)$ exactly or refine it by high-precision Newton iteration, and only then add the Bell-sector transseries. This is not a defect of the expansion; it is a genuine separation of asymptotic levels.

5 A general additive-sector inversion formula

5.1 Reduction to a near-identity map

Let F be locally invertible, let $u = F^{-1}$, and consider

$$G_\varepsilon(x) = F(x) + \varepsilon E(x). \quad (5.1)$$

The bookkeeping parameter ε records the additive sector; it will eventually be set equal to 1. Make the exact change of variable

$$v = F(x), \quad x = u(v), \quad \tilde{E}(v) = E(u(v)).$$

Then solving $G_\varepsilon(x) = Y$ is equivalent to reverting the near-identity equation

$$Y = v + \varepsilon \tilde{E}(v) \quad (5.2)$$

and then applying u . This elementary conjugacy is the source of a closed all-orders formula.

Theorem 5.1 (Lagrange–Buermann formula for an additive perturbation). *Let $F'(u(Y)) \neq 0$. Formally in ε , and analytically whenever the local Lagrange series converges,*

$$G_\varepsilon^{-1}(Y) = u(Y) + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \frac{d^{n-1}}{dY^{n-1}} [u'(Y)E(u(Y))^n]. \quad (5.3)$$

Equivalently, with

$$\mathcal{D}_u = \frac{1}{F'(u)} \frac{d}{du}, \quad (5.4)$$

one has

$$G_\varepsilon^{-1}(Y) = u + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \mathcal{D}_u^{n-1} \left(\frac{E(u)^n}{F'(u)} \right). \quad (5.5)$$

Proof. The generalized Lagrange–Buermann formula applied to Equation (5.2) states that, for any analytic test function H ,

$$H(v) = H(Y) + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \frac{d^{n-1}}{dY^{n-1}} [H'(Y)\tilde{E}(Y)^n].$$

Take $H = u$. Since $\tilde{E}(Y) = E(u(Y))$ and $u'(Y) = 1/F'(u(Y))$, this gives Equation (5.3). Finally,

$$\frac{d}{dY} = \frac{1}{F'(u)} \frac{d}{du} = \mathcal{D}_u,$$

which proves Equation (5.5). The same derivation is valid in formal power-series algebra; see standard treatments of Lagrange inversion such as [5]. $\square$

The first three coefficients, with $f_j = F^{(j)}(u)$ and $E_j = E^{(j)}(u)$, are

$$a_1 = -\frac{E_0}{f_1}, \quad (5.6)$$

$$a_2 = \frac{E_0 E_1}{f_1^2} - \frac{f_2 E_0^2}{2f_1^3}, \quad (5.7)$$

$$a_3 = -\frac{E_0 E_1^2 + \frac{1}{2} E_0^2 E_2}{f_1^3} + \frac{3E_0^2 E_1 f_2}{2f_1^4} + \frac{E_0^3 f_3}{6f_1^4} - \frac{E_0^3 f_2^2}{2f_1^5}. \quad (5.8)$$

These formulas agree with direct coefficient matching, but Equation (5.5) is substantially more efficient at high order.

5.2 A triangular coefficient recurrence

For formalization or computer algebra it is useful to have a recurrence that does not differentiate a large closed expression. Write

$$G_\varepsilon^{-1}(Y) = u + \Delta, \quad \Delta = \sum_{n \geq 1} a_n \varepsilon^n. \quad (5.9)$$

Define the ordinary composition polynomials

$$\mathcal{C}_{n,k}(a_1, a_2, \dots) = [\varepsilon^n] \left(\sum_{j \geq 1} a_j \varepsilon^j \right)^k, \quad \mathcal{C}_{0,0} = 1. \quad (5.10)$$

Expanding $F(u + \Delta) + \varepsilon E(u + \Delta) = F(u)$ gives

$$a_n = -\frac{1}{F'(u)} \left[\sum_{k=2}^n \frac{F^{(k)}(u)}{k!} \mathcal{C}_{n,k} + \sum_{k=0}^{n-1} \frac{E^{(k)}(u)}{k!} \mathcal{C}_{n-1,k} \right]. \quad (5.11)$$

Every term on the right depends only on $a_1, \dots, a_{n-1}$, so the recurrence is triangular. The polynomials $\mathcal{C}_{n,k}$ are ordinary partial Bell polynomials; they should not be confused with the Bell numbers $\mathcal{B}_m$ entering $C(x)$.

5.3 Exponential sectors and an explicit operator product

Suppose now that

$$E(x) = e^{\omega x} C(x), \quad (5.12)$$

where ω is fixed and C is algebraic-logarithmic at infinity. For the gamma envelope, $F'(u) = Y\lambda$. Introduce

$$D_\omega = \frac{d}{du} + \omega. \quad (5.13)$$

The derivative structure forces a pure sector factorization.

Theorem 5.2 (Single-frequency sector factorization). *For $F(u) = Y$, $F'(u) = Y\lambda(u)$, and $E(x) = e^{\omega x} C(x)$, the inverse has the formal expansion*

$$G^{-1}(Y) \sim u(Y) + \sum_{n \geq 1} \frac{e^{n\omega u(Y)}}{Y^n} \Phi_n^{(\omega)}(u(Y)). \quad (5.14)$$

Define, for $1 \leq r \leq n-1$,

$$\mathcal{L}_{n,r}^{(\omega)} H = \frac{H' + (n\omega - r\lambda)H}{\lambda}. \quad (5.15)$$

Then

$$\boxed{\Phi_n^{(\omega)} = \frac{(-1)^n}{n!} \left(\mathcal{L}_{n,n-1}^{(\omega)} \circ \dots \circ \mathcal{L}_{n,1}^{(\omega)} \right) \left(\frac{C^n}{\lambda} \right)}, \quad (5.16)$$

with the empty product understood as the identity for $n = 1$.

Proof. The n -th term of Equation (5.5) starts from

$$\frac{E(u)^n}{F'(u)} = Y^{-1} e^{n\omega u} \frac{C(u)^n}{\lambda(u)}.$$

For any integer $r \geq 1$, direct differentiation gives

$$\mathcal{D}_u (Y^{-r} e^{n\omega u} H(u)) = Y^{-r-1} e^{n\omega u} \frac{H' + (n\omega - r\lambda)H}{\lambda}. \quad (5.17)$$

Applying this identity $n-1$ times proves both the factorization and the operator product. $\square$

The first coefficients are

$$\Phi_1^{(\omega)} = -\frac{C}{\lambda}, \quad (5.18)$$

$$\Phi_2^{(\omega)} = \frac{C D_\omega C}{\lambda^2} - \frac{C^2 P_2}{2\lambda^3}, \quad (5.19)$$

$$\begin{aligned} \Phi_3^{(\omega)} = & -\frac{C(D_\omega C)^2}{\lambda^3} - \frac{C^2 D_\omega^2 C}{2\lambda^3} + \frac{3C^2(D_\omega C)P_2}{2\lambda^4} \\ & + \frac{C^3 P_3}{6\lambda^4} - \frac{C^3 P_2^2}{2\lambda^5}. \end{aligned} \quad (5.20)$$

For fixed ω , the large- u estimates $C = O(u^{-1})$, $\lambda \sim \log u$, and $P_k = O((\log u)^k)$ imply

$$\Phi_n^{(\omega)}(u) = O\left(\frac{u^{-n}}{\log u}\right). \quad (5.21)$$

Thus the grading by powers of Y^{-1} is asymptotically strict even when ω is purely imaginary.

5.4 Multi-frequency grids

Let

$$E(x) = \sum_{\alpha=1}^M e^{\omega_\alpha x} C_\alpha(x). \quad (5.22)$$

Substitution into Equation (5.5) and multinomial expansion of $E(u)^n$ show that the inverse support lies on the grid

$$\left\{ Y^{-|\mathbf{r}|} e^{(\mathbf{r} \cdot \boldsymbol{\omega})u} : \mathbf{r} \in \mathbb{N}^M \setminus \{\mathbf{0}\} \right\}. \quad (5.23)$$

Hence

$$G^{-1}(Y) \sim u(Y) + \sum_{\mathbf{r} \neq \mathbf{0}} Y^{-|\mathbf{r}|} e^{(\mathbf{r} \cdot \boldsymbol{\omega})u(Y)} \Phi_{\mathbf{r}}(u(Y)). \quad (5.24)$$

Order the support first by total degree $|\mathbf{r}|$ and then by any fixed monomial order. Purely imaginary frequencies may produce equal magnitudes or resonant phases, but the factor $Y^{-|\mathbf{r}|}$ still makes the total-degree grading well founded.

5.5 Formal versus analytic validity

For the subfactorial problem the series in the auxiliary amplitude ε is more than a formal device. In the v -coordinate, the map

$$v \longmapsto Y - \varepsilon \tilde{E}(v)$$

is a contraction in a small disk around Y for all bounded ε once Y is sufficiently large, because

$$\tilde{E}'(Y) = \frac{E'(u)}{F'(u)} = O\left(\frac{1}{Y u \log u}\right). \quad (5.25)$$

Thus the Lagrange series converges at $\varepsilon = 1$ for each sufficiently large fixed Y . Its subsequent expansion in powers of u^{-1} , however, is generally only asymptotic. This distinction is useful: the sector expansion in Y^{-1} can converge while the Bell and Stirling coefficient expansions diverge.

6 Inverse transseries for subfactorial continuations

6.1 The branch-indexed complex inverse

Apply Theorem 5.2 with $\omega = \omega_\kappa = (2\kappa + 1)\pi i$ and with C from Equation (2.4).

Theorem 6.1 (All-orders inverse of a subfactorial branch). *For every fixed $\kappa \in \mathbb{Z}$, the large inverse branch satisfying $z_\kappa(Y) - u(Y) \rightarrow 0$ has the nested transseries*

$$z_\kappa(Y) \sim u(Y) + \sum_{n \geq 1} \frac{e^{n\omega_\kappa u(Y)}}{Y^n} \Phi_n^{(\omega_\kappa)}(u(Y)), \quad (6.1)$$

where $u(Y)$ is the exact inverse of $\Gamma(x+1)/e$, and the coefficients are given by Equation (5.16). For each fixed N ,

$$z_\kappa(Y) = u(Y) + \sum_{n=1}^N \frac{e^{n\omega_\kappa u(Y)}}{Y^n} \Phi_n^{(\omega_\kappa)}(u(Y)) + O\left(\frac{Y^{-N-1}u^{-N-1}}{\log u}\right). \quad (6.2)$$

The estimate holds along the positive real Y -axis for fixed κ , and sectorially after the usual branch restrictions.

Proof. The formal expansion and coefficient formula follow directly from Theorem 5.2. For the analytic estimate, the contraction argument following Equation (5.25) gives a convergent amplitude expansion for large Y . The derivative bounds $C^{(j)}(u) = O(u^{-j-1})$, together with fixed ω_κ , imply Equation (5.21) inductively through the operator product. The first omitted sector therefore has the stated order, and the remaining convergent tail is geometrically dominated for large Y . $\square$

The first three sectors are obtained by substituting ω_κ into Equations (5.18) to (5.20). In particular,

$$z_\kappa(Y) = u - \frac{e^{\omega_\kappa u} C}{Y} \frac{C}{\lambda} + \frac{e^{2\omega_\kappa u}}{Y^2} \left[\frac{C(C' + \omega_\kappa C)}{\lambda^2} - \frac{C^2(\lambda^2 + \lambda')}{2\lambda^3} \right] + O\left(\frac{Y^{-3}u^{-3}}{\log u}\right). \quad (6.3)$$

For the principal incomplete-gamma path, $\kappa = 0$ and $\omega_0 = \pi i$.

6.2 Expanding the first Bell sector

Let

$$L = \log u. \quad (6.4)$$

The digamma expansion [2] gives

$$\lambda = \psi(u+1) \sim L + \frac{1}{2u} - \frac{1}{12u^2} + \frac{1}{120u^4} - \cdots. \quad (6.5)$$

Combining this with Equation (2.15) gives

$$\frac{C(u)}{\lambda(u)} \sim \frac{1}{uL} - \frac{4L+1}{2u^2L^2} + \frac{60L^2+13L+3}{12u^3L^3}$$

$$- \frac{360L^3 + 64L^2 + 14L + 3}{24u^4L^4} + O\left(\frac{u^{-5}}{L}\right). \quad (6.6)$$

At the next sector,

$$\Phi_2^{(\omega)}(u) = \frac{1}{u^2} \left(-\frac{1}{2L} + \frac{\omega}{L^2} \right) + O\left(\frac{u^{-3}}{L}\right). \quad (6.7)$$

Consequently, the branch inverse begins

$$\begin{aligned} z_\kappa(Y) = u - \frac{e^{\omega_\kappa u}}{Y} & \left[\frac{1}{uL} - \frac{4L+1}{2u^2L^2} + \frac{60L^2+13L+3}{12u^3L^3} + O\left(\frac{u^{-4}}{L}\right) \right] \\ & + \frac{e^{2\omega_\kappa u}}{Y^2} \left[-\frac{1}{2u^2L} + \frac{\omega_\kappa}{u^2L^2} + O\left(\frac{u^{-3}}{L}\right) \right] + O\left(\frac{Y^{-3}u^{-3}}{L}\right). \end{aligned} \quad (6.8)$$

All coefficients at all orders are generated by Bell numbers from C , Bernoulli numbers from λ , and the differential operators $\mathcal{L}_{n,r}^{(\omega)}$.

6.3 The real symmetrized inverse

Set

$$H(u) = C(u) \cos(\pi u), \quad H'(u) = C'(u) \cos(\pi u) - \pi C(u) \sin(\pi u). \quad (6.9)$$

Applying Equation (5.5) with $E = H$ yields a real expansion.

Theorem 6.2 (Real inverse interpolating every derangement). *The eventual real inverse $r(Y) = \mathfrak{D}_{\text{sym}}^{-1}(Y)$ satisfies*

$$r(Y) = u - \frac{H}{Y\lambda} + \frac{1}{Y^2} \left[\frac{HH'}{\lambda^2} - \frac{H^2(\lambda^2 + \lambda')}{2\lambda^3} \right] + O\left(\frac{Y^{-3}u^{-3}}{\log u}\right). \quad (6.10)$$

More generally,

$$r(Y) \sim u + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{1}{Y\lambda} \frac{d}{du} \right)^{n-1} \left(\frac{H(u)^n}{Y\lambda} \right). \quad (6.11)$$

Successive orders contain harmonics $e^{im\pi u}$ with $m = -n, -n+2, \dots, n$.

The first-order displacement is

$$r(Y) - u(Y) \sim -\frac{\cos(\pi u)}{Y u \log u}. \quad (6.12)$$

At the exact derangement ordinates, $r(D_n) = n$.

6.4 Parity-resolved inverse transseries

For $G_\sigma = F + \sigma C$, set $\omega = 0$ and use the amplitude $\varepsilon = \sigma$. Then

$$\boxed{x_\sigma(Y) \sim u(Y) + \sum_{n \geq 1} \frac{\sigma^n}{Y^n} \Phi_n^{(0)}(u(Y))}. \quad (6.13)$$

The first two sectors are

$$x_\sigma(Y) = u - \frac{\sigma C}{Y\lambda} + \frac{1}{Y^2} \left[\frac{CC'}{\lambda^2} - \frac{C^2(\lambda^2 + \lambda')}{2\lambda^3} \right] + O\left(\frac{Y^{-3}u^{-3}}{\log u}\right). \quad (6.14)$$

Expanding the coefficient functions gives

$$\begin{aligned} x_\sigma(Y) = u - \frac{\sigma}{Y} & \left[\frac{1}{uL} - \frac{4L+1}{2u^2L^2} + \frac{60L^2+13L+3}{12u^3L^3} + O\left(\frac{u^{-4}}{L}\right) \right] \\ & + \frac{1}{Y^2} \left[-\frac{1}{2u^2L} + \frac{8L^2-3L-2}{4u^3L^3} + O\left(\frac{u^{-4}}{L}\right) \right] + O\left(\frac{Y^{-3}u^{-3}}{L}\right). \end{aligned} \quad (6.15)$$

This is the simplest explicit real form of the inverse-subfactorial transseries.

6.5 The alternating displacement at derangement ordinates

Let $\sigma_n = (-1)^n$, and let $u_n = u(D_n)$. Since $D_n = F(n) + \sigma_n C(n)$, Taylor inversion of F around n gives

$$u_n - n = \frac{\sigma_n C(n)}{F(n)\psi(n+1)} + O\left(\frac{1}{F(n)^2 n^2 \log n}\right). \quad (6.16)$$

Therefore

$$\boxed{u(D_n) - n \sim \frac{(-1)^n}{D_n n \log n}}. \quad (6.17)$$

The inverse-gamma envelope passes alternately just above the even indices and just below the odd indices when sampled at the exact derangement values. The Bell sector in Equation (6.15) cancels this displacement and recovers the integer exactly, order by order.

6.6 Complex branch bookkeeping beyond the principal real carrier

The positive real inverse uses the principal logarithm and W_0 . In the complex plane, write

$$A_m = \text{Log} \frac{eY}{\sqrt{2\pi}} + 2\pi im, \quad m \in \mathbb{Z}, \quad (6.18)$$

and choose a Lambert branch $W_j(A_m/e)$. Every pair (m, j) for which $Q_{m,j} = A_m/W_j(A_m/e)$ tends to infinity inside a Stirling sector generates a formal inverse-gamma branch by Theorem 4.1. The incomplete-gamma continuation contributes the additional branch label κ through ω_κ . Thus the full complex branch structure is naturally organized by a logarithm index, a Lambert index, and an incomplete-gamma path index. The present article focuses on $(m, j) = (0, 0)$ and arbitrary fixed κ , which is the family asymptotic to the positive real envelope.

7 General theory for transseries of the same shape

7.1 Two-stage inversion architecture

The subfactorial problem separates into two logically independent reversions.

Stage A: invert the dominant carrier

Suppose a shifted coordinate $q = x + \beta$ reduces the logarithm of the dominant function to

$$\log F(x) = c + q(\log q - a) + R(q), \quad (7.1)$$

where $R(q) = o(q)$ belongs to a differential algebra of inverse powers and logarithms. For $S = \log Y - c$, use

$$w = W_0(Se^{-a}), \quad Q = S/w = e^{a+w} \quad (7.2)$$

as the exact Lambert carrier. Revert the smaller term R by the triangular recurrence of Theorem 4.1. The universal reversion denominator is $1 + w$.

Stage B: invert the subdominant sectors

Write the full function as

$$G(x) = F(x) + E(x), \quad (7.3)$$

where $E/F \rightarrow 0$. Keep $u = F^{-1}(Y)$ exact and apply the additive Lagrange–Bürmann formula

$$G^{-1}(Y) \sim u + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{1}{F'(u)} \frac{d}{du} \right)^{n-1} \left(\frac{E(u)^n}{F'(u)} \right). \quad (7.4)$$

Only after this sector expansion is established should one substitute the Stage A expansion for u .

The architecture prevents a common error: flattening every scale too early can hide a tiny sector beneath the truncation error of a much larger divergent expansion.

7.2 Lambert-reducible polynomial-logarithmic corrections

The recurrence in Theorem 4.1 extends directly when

$$R(q) \sim \sum_{m \geq m_0} q^{-m} P_m(\log q), \quad (7.5)$$

with polynomial or Laurent-polynomial P_m . Under $q = Q(1 + U)$, one expands

$$\log q = \log Q + \log(1 + U)$$

automatically. The coefficient ring must contain $\log Q = a + w$ and the reversion factor $s^{-1} = (1 + w)^{-1}$, but the recursion remains triangular because $U = O(Q^{-1-m_0})$. More general well-based logarithmic transseries can be handled in the same way whenever composition with $\log(1 + U)$, differentiation, and division by $1 + w$ remain inside the chosen coefficient field. This is the operational setting of grid-based transseries described in standard references [6, 7].

At first order, the correction has the universal Newton form

$$q - Q \sim -\frac{R(Q)}{1 + w}. \quad (7.6)$$

Higher orders are forced by repeated composition. The same denominator $1 + w$ appears regardless of the constant a in $q(\log q - a)$, because

$$\frac{d}{dq} [q(\log q - a)] = \log q - a + 1 = w + 1$$

at the carrier point.

7.3 Moment/Stieltjes corrections

The Bell correction belongs to a much larger class with an especially transparent remainder theory.

Theorem 7.1 (Moment expansion for a Stieltjes correction). *Let μ be a positive measure on $[0, \infty)$ with finite moments*

$$M_j = \int_0^\infty t^j d\mu(t), \quad j = 0, 1, \dots, M. \quad (7.7)$$

For

$$C_\mu(x) = \int_0^\infty \frac{d\mu(t)}{x+t}, \quad x > 0, \quad (7.8)$$

one has the exact finite expansion

$$C_\mu(x) = \sum_{k=1}^M (-1)^{k-1} \frac{M_{k-1}}{x^k} + R_{\mu,M}(x), \quad (7.9)$$

where

$$R_{\mu,M}(x) = (-1)^M x^{-M} \int_0^\infty \frac{t^M}{x+t} d\mu(t). \quad (7.10)$$

In particular,

$$(-1)^M R_{\mu,M}(x) \geq 0, \quad |R_{\mu,M}(x)| \leq M_M x^{-M-1}. \quad (7.11)$$

Proof. Integrate the exact resolvent identity Equation (2.14) against μ . $\square$

Now consider

$$G(x) = F(x) + e^{\omega x} C_\mu(x). \quad (7.12)$$

The inverse transseries is still Equation (5.14). The first sector is

$$-\frac{e^{\omega u}}{Y\lambda(u)} \sum_{k \geq 1} (-1)^{k-1} \frac{M_{k-1}}{u^k}, \quad (7.13)$$

followed by corrections generated by differentiation and products of the moments. Thus a moment sequence in the original additive term becomes the coefficient sequence of a new inverse transseries sector. The Bell-number phenomenon is the special case $M_{k-1} = \mathcal{B}_k$.

7.4 Multi-frequency and resonant corrections

For

$$G(x) = F(x) + \sum_{\alpha=1}^M e^{\omega_\alpha x} C_\alpha(x), \quad (7.14)$$

the inverse support is the additive semigroup generated by

$$\mathfrak{m}_\alpha(Y) = \frac{e^{\omega_\alpha u(Y)}}{Y}. \quad (7.15)$$

Mixed products produce $\mathbf{m}_1^{r_1} \cdots \mathbf{m}_M^{r_M}$. If a resonance $\mathbf{r} \cdot \boldsymbol{\omega} = \mathbf{s} \cdot \boldsymbol{\omega}$ occurs, the corresponding coefficient functions add. Resonance changes coefficient bookkeeping but not the total-degree valuation $Y^{-|\mathbf{r}|}$.

The real symmetrized subfactorial is the two-frequency example $\omega_{\pm} = \pm\pi i$. At total degree n , its harmonics have frequencies $(n - 2j)\pi$, $0 \leq j \leq n$. This explains the harmonic statement in Theorem 6.2 without separate trigonometric calculations at every order.

7.5 Multiplicative perturbations

If

$$G(x) = F(x) (1 + M(x)), \quad M(x) \rightarrow 0, \quad (7.16)$$

then take $E(x) = F(x)M(x)$ in Equation (7.4). Since $E(u) = YM(u)$, the first inverse displacement is

$$G^{-1}(Y) - u(Y) \sim -\frac{M(u)}{\lambda(u)}. \quad (7.17)$$

Thus the inverse-sector monomial is the relative correction $M(u)$, not Y^{-1} . The same Lagrange–Buermann operator handles additive, multiplicative, oscillatory, and mixed perturbations; only the valuation changes.

7.6 Closure of the coefficient field

For the subfactorial problem, let $L = \log u$ and consider the differential algebra

$$\mathcal{A} = \mathbb{C}[\omega, L, L^{-1}] \left[\left[u^{-1} \right] \right]. \quad (7.18)$$

Because

$$\frac{d}{du} (u^{-m} L^j) = -m u^{-m-1} L^j + j u^{-m-1} L^{j-1},$$

the algebra is closed under differentiation. The expansion $\lambda = L + O(u^{-1})$ makes λ a unit after adjoining L^{-1} , so division by λ is also closed. Since $C \in u^{-1}\mathcal{A}$, the operator product Equation (5.16) proves inductively that

$$\Phi_n^{(\omega)} \in u^{-n} L^{-1} \mathcal{A}. \quad (7.19)$$

The full single-frequency inverse belongs to the grid extension

$$u + \mathcal{A}[[\mathbf{m}]], \quad \mathbf{m} = Y^{-1} e^{\omega u}. \quad (7.20)$$

This gives a precise algebraic meaning to the nested inverse transseries used throughout the article.

7.7 Residual-to-error transfer

A formal expansion becomes a certified approximation only after its residual is divided by a derivative lower bound.

Lemma 7.2 (Real residual certificate). *Let G be continuously differentiable and strictly monotone on an interval containing an exact root x_* of $G(x) = Y$ and an approximation x_N . If*

$$|G'(x)| \geq m_N > 0 \quad (7.21)$$

throughout the segment joining x_N and x_* , then

$$|x_N - x_*| \leq \frac{|G(x_N) - Y|}{m_N}. \quad (7.22)$$

Proof. Apply the mean-value theorem to $G(x_N) - G(x_*)$. $\square$

For the real subfactorial interpolations and large u ,

$$G'(x) = F(x)\psi(x+1) + O(x^{-1}).$$

On a sufficiently small neighborhood of u , one may take

$$m_N \geq \frac{1}{2}Y \log u. \quad (7.23)$$

Therefore a computable residual yields the practical certificate

$$|x_N - x_*| \leq \frac{2|G(x_N) - Y|}{Y \log u}. \quad (7.24)$$

For complex branches, the corresponding tool is a local inverse-function or Rouché argument on a disk; the same derivative dominance supplies the needed nonvanishing condition.

7.8 A reusable inversion algorithm

Algorithm 7.3 (Nested transseries inversion). For a gamma-dominant function $G = F + E$ and a large target Y :

1. Identify a shift $q = x + \beta$ that reduces $\log F$ to $c + q(\log q - a) + R(q)$.
2. Form $S = \log Y - c$, $w = W_0(Se^{-a})$, and $Q = S/w$.
3. Generate as many Stage A coefficients as needed from Equation (4.15). Use the resulting approximation as a starting value for Newton iteration on $\log F(x) - \log Y$, thereby obtaining the exact carrier u at the precision required by the smallest desired sector.
4. Evaluate $E(u)$, $F'(u)$, and their derivatives. For Stieltjes corrections, either use the convergent integral/partial-fraction representation or optimally truncate the moment expansion with its explicit remainder bound.
5. Generate Stage B coefficients by Equation (5.5), the recurrence Equation (5.11), or the operator product Equation (5.16).
6. Sum sectors in increasing valuation and compute the residual $G(x_N) - Y$.
7. Convert the residual to an error bound with Equation (7.22) or a complex inverse-function estimate.

Using $\log F$ in the Newton step avoids overflow. The required working precision must be chosen from the smallest requested absolute correction. In the subfactorial case, resolving the first Bell sector at large Y requires roughly the number of decimal digits in Y , in addition to the digits desired in the answer.

8 A Touchard-polynomial deformation

The Bell numbers arise because the correction measure is Poisson with parameter 1, shifted by one. Replacing the fixed incomplete-gamma argument -1 by $-a$ produces the full Touchard-polynomial family.

8.1 Generalized alternating exponential truncations

Fix $a > 0$ and define

$$D_n(a) = n! \sum_{j=0}^n \frac{(-a)^j}{j!} = e^{-a} \Gamma(n+1, -a). \quad (8.1)$$

For a path index κ , put

$$\omega_{\kappa,a} = \log a + (2\kappa + 1)\pi i \quad (8.2)$$

and

$$C_a(z) = ae^{-a} \int_0^1 t^z e^{at} dt = ae^{-a} \sum_{j=0}^{\infty} \frac{a^j}{j!(z+j+1)}. \quad (8.3)$$

The same path calculation as in [Theorem 2.1](#) gives

$$\boxed{\mathfrak{D}_{\kappa,a}(z) = e^{-a} \Gamma(z+1) + e^{\omega_{\kappa,a}z} C_a(z).} \quad (8.4)$$

At integer n , the phase is $(-a)^n$, and [Equation \(8.4\)](#) equals $D_n(a)$.

8.2 Touchard coefficients and exact remainder

Let $\mathcal{T}_m(a)$ be the Touchard polynomial, equivalently the m -th moment of a Poisson random variable of mean a :

$$\mathcal{T}_m(a) = e^{-a} \sum_{j=0}^{\infty} \frac{j^m a^j}{j!}. \quad (8.5)$$

A shift of index gives

$$ae^{-a} \sum_{j=0}^{\infty} \frac{a^j (j+1)^{m-1}}{j!} = \mathcal{T}_m(a). \quad (8.6)$$

Therefore [Theorem 7.1](#) yields the following direct extension of the Bell theorem.

Theorem 8.1 (Touchard expansion). *For every $x > 0$ and integer $M \geq 1$,*

$$C_a(x) = \sum_{m=1}^M (-1)^{m-1} \frac{\mathcal{T}_m(a)}{x^m} + R_{a,M}(x), \quad (8.7)$$

where

$$(-1)^M R_{a,M}(x) > 0, \quad |R_{a,M}(x)| \leq \frac{\mathcal{T}_{M+1}(a)}{x^{M+1}}. \quad (8.8)$$

At $a = 1$, $\mathcal{T}_m(1) = \mathcal{B}_m$, recovering [Theorem 2.5](#).

8.3 Inverse transseries for the deformed family

Let $u_a(Y)$ solve

$$e^{-a}\Gamma(u_a + 1) = Y. \quad (8.9)$$

Its Lambert carrier is obtained by replacing A with

$$A_a = \log \frac{e^a Y}{\sqrt{2\pi}}. \quad (8.10)$$

All centered Stirling coefficients remain unchanged. Applying Theorem 5.2 gives

$$\mathfrak{D}_{\kappa,a}^{-1}(Y) \sim u_a(Y) + \sum_{n \geq 1} \frac{e^{n\omega_{\kappa,a}u_a(Y)}}{Y^n} \Phi_{a,n}^{(\omega_{\kappa,a})}(u_a(Y)), \quad (8.11)$$

where $\Phi_{a,n}$ is obtained from Equation (5.16) by replacing C with C_a . In particular,

$$\mathfrak{D}_{\kappa,a}^{-1}(Y) = u_a(Y) - \frac{e^{\omega_{\kappa,a}u_a}C_a(u_a)}{Y\psi(u_a + 1)} + O\left(\frac{|a|^{2u_a}}{Y^2u_a^2 \log u_a}\right). \quad (8.12)$$

For fixed a , factorial growth still dominates a^{u_a} , so the sector is asymptotically small. This family demonstrates that Bell-number inverse sectors are one point in a broader Poisson-moment/Touchard theory.

9 Numerical verification

The numerical experiments in this section use arbitrary-precision arithmetic. The exact carrier $u(Y)$ was obtained by Newton iteration on

$$\log \Gamma(u + 1) - 1 - \log Y = 0, \quad (9.1)$$

seeded by Equation (4.26). The correction $C(u)$ and its derivative were evaluated from the rapidly convergent sums

$$C(u) = \frac{1}{e} \sum_{j \geq 0} \frac{1}{j!(u + j + 1)}, \quad C'(u) = -\frac{1}{e} \sum_{j \geq 0} \frac{1}{j!(u + j + 1)^2}. \quad (9.2)$$

All displayed differences are signed errors, rounded from computations performed with at least 80 decimal digits.

9.1 Accuracy of the Lambert–Stirling carrier

Let $u^{[0]} = Q - 1/2$, and let $u^{[1]}, u^{[3]}, u^{[5]}, u^{[7]}$ denote the successive truncations obtained by adding $d_1Q^{-1}, d_3Q^{-3}, d_5Q^{-5}, d_7Q^{-7}$. At the target $Y = D_n$, the errors relative to the exact gamma-envelope inverse $u(D_n)$ are as follows.

The alternating decrease is consistent with the centered Stirling structure. More importantly, at $n = 20$ the Bell displacement is only about 1.7×10^{-20} , already much smaller than the Q^{-7} -truncation error. At $n = 50$ the difference is overwhelming. This is the practical manifestation of the scale separation in Equation (4.32): one must Newton-refine the carrier before attempting to resolve the subfactorial sector.

Table 1: Signed errors $u^{[j]} - u(D_n)$ in the Lambert–Stirling carrier.

n	$u^{[0]} - u$	$u^{[1]} - u$	$u^{[3]} - u$	$u^{[5]} - u$	$u^{[7]} - u$
10	$-1.6867983 \times 10^{-3}$	1.2192504×10^{-6}	$-3.3010676 \times 10^{-9}$	$2.0997717 \times 10^{-11}$	$-2.5455408 \times 10^{-13}$
20	$-6.7283562 \times 10^{-4}$	1.1908008×10^{-7}	$-8.5346644 \times 10^{-11}$	$1.4599668 \times 10^{-13}$	$-4.7355933 \times 10^{-16}$
50	$-2.1036963 \times 10^{-4}$	5.7995124×10^{-9}	$-6.9095404 \times 10^{-13}$	$1.9819752 \times 10^{-16}$	$-1.0724457 \times 10^{-19}$

9.2 Recovery of integer indices from exact derangement values

For $Y = D_n$, choose $\sigma = (-1)^n$. Let

$$x_\sigma^{[1]} = u - \frac{\sigma C}{Y\lambda}, \quad (9.3)$$

$$x_\sigma^{[2]} = x_\sigma^{[1]} + \frac{1}{Y^2} \left[\frac{CC'}{\lambda^2} - \frac{C^2(\lambda^2 + \lambda')}{2\lambda^3} \right]. \quad (9.4)$$

Because $x_{(-1)^n}(D_n) = n$ exactly, their signed errors may be measured directly against n .

 Table 2: Bell-sector recovery of the integer index from $Y = D_n$.

n	$u(D_n) - n$	$x_{(-1)^n}^{[1]}(D_n) - n$	$x_{(-1)^n}^{[2]}(D_n) - n$
10	$2.67167149055 \times 10^{-8}$	$9.13998505805 \times 10^{-16}$	$4.21393368757 \times 10^{-23}$
11	$-2.15649411782 \times 10^{-9}$	$6.12409529877 \times 10^{-18}$	$-2.34134360394 \times 10^{-26}$
20	$1.68471784606 \times 10^{-20}$	$4.43896764784 \times 10^{-40}$	$1.56737019375 \times 10^{-59}$
21	$-7.55330911992 \times 10^{-22}$	$9.04443664056 \times 10^{-43}$	$-1.45092932399 \times 10^{-63}$

The sign of $u(D_n) - n$ alternates exactly as predicted by Equation (6.17). Each added sector contributes another power of the tiny amplitude Y^{-1} ; in the displayed cases this roughly doubles the number of correct decimal places at each step.

9.3 A genuinely complex inverse branch

For the principal incomplete-gamma branch $\omega = \pi i$, define $z^{[1]}$ and $z^{[2]}$ by retaining one and two sectors in Equation (6.3). The next table compares them with a high-precision numerical root of $\mathfrak{D}_0(z) = Y$.

Table 3: Principal complex branch: displacement and truncation errors.

Y	$z_0(Y) - u(Y)$	errors $z^{[1]} - z_0, z^{[2]} - z_0$
10^8	$-2.2116333803 \times 10^{-10}$	$-2.4490805670 \times 10^{-19}$
	$1.8899727089 \times 10^{-10}i$	$1.5375833604 \times 10^{-19}i$
		$-1.0624249085 \times 10^{-28}$
		$4.2279221738 \times 10^{-28}i$
10^{20}	$-1.8355790924 \times 10^{-23}$	$-4.4860789094 \times 10^{-44}$
	$1.3633601642 \times 10^{-22}i$	$4.9345285404 \times 10^{-44}i$
		$4.7644684050 \times 10^{-65}$
		$1.1986677056 \times 10^{-65}i$

The phase $e^{\pi i u}$ is visible in the nonzero imaginary part. The error reduction agrees with the sector orders Y^{-2} and Y^{-3} .

9.4 A residual check

The most reliable implementation should not trust a coefficient table alone. Given any approximant x_N , evaluate

$$\text{Res}_N = G(x_N) - Y.$$

For the real parity or symmetrized interpolation, Equation (7.24) converts this directly into a certified error. For example, once Y is large enough that $G'(x) \geq \frac{1}{2}Y \log u$ on the enclosing interval, a residual of size $10^{-p}Y \log u$ guarantees p correct absolute decimal digits in x_N . This residual check also detects branch mistakes: using $e^{-\pi i u}$ instead of $e^{\pi i u}$ generally leaves a first-sector residual rather than a third-sector residual.

10 Consequences, extensions, and research directions

10.1 A concise hierarchy theorem

Combining the preceding results gives the following summary.

Theorem 10.1 (Hierarchy of the inverse subfactorial). *Let $Y \rightarrow +\infty$, let A, w, Q, s be as in Equations (4.5) and (4.8), and let $u = F^{-1}(Y)$. Then:*

1. $u \sim Q - 1/2$, and the complete algebraic correction is a series in odd powers of Q^{-1} with coefficients rational in s .
2. The principal analytic subfactorial inverse is

$$\mathfrak{D}_0^{-1}(Y) = u - \frac{e^{\pi i u} C(u)}{Y \psi(u+1)} + O\left(\frac{Y^{-2} u^{-2}}{\log u}\right).$$

3. The real symmetrized inverse is

$$\mathfrak{D}_{\text{sym}}^{-1}(Y) = u - \frac{C(u) \cos(\pi u)}{Y \psi(u+1)} + O\left(\frac{Y^{-2} u^{-2}}{\log u}\right).$$

4. The parity inverse is

$$x_\sigma(Y) = u - \frac{\sigma C(u)}{Y \psi(u+1)} + O\left(\frac{Y^{-2} u^{-2}}{\log u}\right).$$

5. The discrete generalized inverse is $N(Y) = \left\lceil \mathfrak{D}_{\text{sym}}^{-1}(Y) \right\rceil$ for all sufficiently large Y .

In every smooth case, all higher sectors are generated by Equation (5.5); in the analytic branch case they factor as $Y^{-n} e^{n\pi i u} \Phi_n(u)$.

10.2 Why the Bell numbers survive inversion

The Bell numbers enter the original sequence additively, not relatively. It might therefore seem that inversion would destroy their simple structure. The first sector shows the opposite: inversion multiplies the Bell expansion by the reciprocal logarithmic derivative $1/\psi(u+1)$. At higher orders, the operator product differentiates and multiplies copies of C , so the coefficients are finite polynomial expressions in Bell numbers and Bernoulli numbers. In other words, inversion does not erase the combinatorial moments; it promotes them into a differential algebra.

At total sector degree n , the leading algebraic size is $u^{-n}/\log u$. The coefficient of a fixed power u^{-m} is assembled from compositions of m into n positive parts, weighted by products of Bell numbers and by the normalized gamma derivatives P_k . This gives a natural combinatorial interpretation of the inverse coefficients in terms of two interacting Bell structures: ordinary Bell numbers from the Poisson moment measure and complete exponential Bell polynomials from derivatives of $F = \exp(\log F)$.

10.3 Hyperasymptotic flattening

The nested answer deliberately leaves $u(Y)$ exact. A fully flattened transseries that resolves the Bell sector using only explicit elementary, Lambert, and exponential monomials would have to continue the inverse-gamma expansion through its resurgent hierarchy. Because the Bell scale is smaller than e^{-cQ} for every fixed c , finitely many exponential improvements of Stirling’s series still do not suffice asymptotically. One would need either:

- an exact special-function carrier, as used here;
- a hyperasymptotic process whose exponential level increases with Q ; or
- a convergent representation of $\log \Gamma$, such as a Binet-type integral, incorporated into the inversion.

Developing a practically efficient “all-the-way-down” flattened transseries is an interesting problem because it couples resurgence with a superexponentially small combinatorial sector.

10.4 Global complex branch geometry

The local large branch near the positive envelope is unambiguous, but the global inverse of $\Gamma(z+1, -1)$ has a much richer Riemann-surface structure. Critical points satisfy

$$F(z)\psi(z+1) + e^{\omega_\kappa z}(C'(z) + \omega_\kappa C(z)) = 0. \quad (10.1)$$

Near the positive axis the first term dominates, so no critical point interferes with the branch constructed here. Farther into the complex plane, zeros of ψ , gamma oscillation, and exponential phase growth can compete. A systematic classification would combine the indices (m, j, κ) of Equation (6.18) with Stokes geometry for the gamma function.

10.5 The staircase and periodic transseries

Equation (3.13) places the discrete inverse in a different category from well-based small transseries: its periodic correction is order one and discontinuous. Nonetheless, one can regard it

as a transasymptotic composition

$$N(Y) = \mathcal{S}(r(Y)), \quad \mathcal{S}(x) = [x],$$

where the smooth argument $r(Y)$ has the complete expansion developed here. Questions about the distribution of the fractional part of $r(Y)$, or about averaged versions of $N(Y) - r(Y)$, may be approachable by exponential-sum methods. The rapidly increasing gaps $D_{n+1} - D_n$ suggest that averaging with respect to Y and averaging with respect to n will lead to different limiting laws.

10.6 Formalization prospects

The derivation separates into components well suited to proof-assistant formalization:

1. the exact incomplete-gamma decomposition [Theorem 2.1](#);
2. the finite resolvent identity and Bell remainder theorem [Theorem 2.5](#);
3. the formal Lambert-centered recurrence [Theorem 4.1](#);
4. the formal Lagrange–Buermann identity [Theorem 5.1](#);
5. coefficient extraction in a differential ring;
6. residual-to-error transfer by [Theorem 7.2](#).

The analytic Stirling remainder and sectorial complex estimates are the least elementary pieces. A practical formalization strategy would first prove the formal coefficient theorems and the real finite-order error statements, leaving full complex resurgence for a later layer.

Problem 10.2 (Coefficient combinatorics). Find a direct closed formula for the coefficient of $u^{-m}L^{-j}$ in $\Phi_n^{(0)}$ in terms of set partitions, compositions, Bell numbers $\mathcal{B}_k$, and Bernoulli numbers. The operator product proves existence and gives an algorithm, but a transparent combinatorial formula may reveal positivity or sign regularities not visible from differentiation.

Problem 10.3 (Uniform Touchard deformation). Study [Equation \(8.11\)](#) when $a = a(Y)$ grows with Y . The fixed- a hierarchy breaks when a^{u_a}/Y ceases to be small. Transition regimes should occur when $\log a$ is comparable with $\log u_a - 1$, and may require a new two-parameter Lambert carrier.

Problem 10.4 (Convergence boundary in the amplitude plane). For fixed large Y , determine the nearest singularity in ε of the solution to $F(x) + \varepsilon e^{\omega x} C(x) = Y$. The Lagrange series converges at $\varepsilon = 1$ for sufficiently large Y , but the precise radius is controlled by simultaneous solutions of the equation and its x -derivative. Its asymptotic expansion would quantify the true convergence margin of the Bell-sector series.

11 Conclusion

The inverse subfactorial is not a single object until one specifies how the integer sequence is continued or inverted. Once that choice is made, the asymptotic structure is remarkably systematic.

The dominant factorial growth is inverted by a Lambert W carrier. Centered Stirling corrections are then reversed through a triangular algebra whose universal denominator is $1 + W$. The exact derangement correction is a Stieltjes transform whose moments are Bell numbers, and additive

Lagrange–Buermann inversion promotes it to a new sector graded by Y^{-1} . For a branch of the incomplete gamma continuation, the sector carries the phase $e^{(2\kappa+1)\pi i u}$; for the real symmetrization, the phases combine into Fourier harmonics; for parity envelopes, they reduce to signs; and for the discrete inverse, a separate order-one ceiling sector remains.

The central all-orders result is the nested formula

$$\mathfrak{D}_\kappa^{-1}(Y) \sim u(Y) + \sum_{n \geq 1} \frac{e^{n(2\kappa+1)\pi i u(Y)}}{Y^n} \Phi_n^{(\kappa)}(u(Y)),$$

with $u(Y)$ given by the Lambert-centered inverse-gamma expansion and Φ_n given by a finite differential-operator product. The same two-stage method applies to a broad class of gamma-dominant maps with polynomial-logarithmic carriers and moment-generated additive sectors. The Touchard deformation shows that the Bell coefficients are not accidental: they are the Poisson-parameter-one instance of a general moment-to-inverse principle.

A Coefficient extraction for the centered gamma inverse

This appendix records the algebra behind Equations (4.22) to (4.25). Write

$$R(q) = r_1 q^{-1} + r_2 q^{-2} + r_3 q^{-3} + \dots \quad (\text{A.1})$$

and

$$q = Q + d_1 Q^{-1} + d_2 Q^{-2} + d_3 Q^{-3} + \dots \quad (\text{A.2})$$

Substitution into Equation (4.14) gives, successively,

$$s d_1 + r_1 = 0, \quad (\text{A.3})$$

$$s d_2 + r_2 = 0, \quad (\text{A.4})$$

$$s d_3 + r_3 - r_1 d_1 + \frac{1}{2} d_1^2 = 0, \quad (\text{A.5})$$

$$s d_4 + r_4 - 2 r_1 d_2 - r_2 d_1 + d_1 d_2 = 0. \quad (\text{A.6})$$

Using $d_1 = -r_1/s$ and $d_2 = -r_2/s$ yields Equations (4.20) and (4.21). At the next order one obtains

$$\begin{aligned} d_5 = & -\frac{r_5}{s} - \frac{4r_1 r_3 + 2r_2^2}{s^2} - \frac{2r_1^3 + r_1 r_3 + \frac{1}{2} r_2^2}{s^3} \\ & - \frac{5r_1^3}{3s^4} - \frac{r_1^3}{2s^5}. \end{aligned} \quad (\text{A.7})$$

For the centered gamma series,

$$r_1 = -\frac{1}{24}, \quad r_2 = 0, \quad r_3 = \frac{7}{2880}, \quad r_4 = 0, \quad r_5 = -\frac{31}{40320}.$$

Substitution into Equation (A.7) simplifies to Equation (4.24).

The vanishing of all d_{2m} follows by parity. If $U(t)$ contains only even powers of t , then every term in the right-hand side of Equation (4.14) contains only odd powers: $t^{-1}U^k$ is odd and $t^{2m-1}(1+U)^{-(2m-1)}$ is odd. Therefore no even coefficient equation is generated, and the unique solution has $d_{2m} = 0$.

B Derivation of the first Bell-log coefficients

From Equations (2.15) and (6.5), write

$$\begin{aligned} C(u) &= u^{-1} - 2u^{-2} + 5u^{-3} - 15u^{-4} + O(u^{-5}), \\ \lambda(u) &= L + \frac{1}{2}u^{-1} - \frac{1}{12}u^{-2} + O(u^{-4}). \end{aligned}$$

Factor $\lambda = L(1 + \eta)$, where

$$\eta = \frac{1}{2uL} - \frac{1}{12u^2L} + O(u^{-4}L^{-1}).$$

Then

$$\begin{aligned} \lambda^{-1} &= L^{-1}(1 - \eta + \eta^2 - \eta^3 + \dots) \\ &= \frac{1}{L} - \frac{1}{2uL^2} + \frac{1}{u^2} \left(\frac{1}{12L^2} + \frac{1}{4L^3} \right) + O\left(\frac{u^{-3}}{L^2}\right). \end{aligned}$$

Multiplication by the Bell expansion of C gives

$$\begin{aligned} \frac{C}{\lambda} &= \frac{1}{uL} + \frac{1}{u^2} \left(-\frac{2}{L} - \frac{1}{2L^2} \right) \\ &\quad + \frac{1}{u^3} \left(\frac{5}{L} + \frac{13}{12L^2} + \frac{1}{4L^3} \right) + O\left(\frac{u^{-4}}{L}\right), \end{aligned}$$

which is equivalent to the first three terms of Equation (6.6).

For $\omega = 0$,

$$\Phi_2^{(0)} = \frac{CC'}{\lambda^2} - \frac{C^2}{2\lambda} - \frac{C^2\lambda'}{2\lambda^3}. \quad (\text{B.1})$$

Straight expansion yields

$$\Phi_2^{(0)} = -\frac{1}{2u^2L} + \frac{8L^2 - 3L - 2}{4u^3L^3} + O\left(\frac{u^{-4}}{L}\right). \quad (\text{B.2})$$

For general fixed ω , add $\omega C^2/\lambda^2$, giving Equation (6.7).

C A second proof of the additive coefficient formula

For completeness, we verify directly that Equation (5.5) solves the inverse problem. Substitute

$$x = u + \sum_{n \geq 1} a_n \varepsilon^n$$

into

$$F(x) + \varepsilon E(x) = F(u).$$

At order ε ,

$$F'a_1 + E = 0,$$

so $a_1 = -E/F'$. At order ε^2 ,

$$F'a_2 + \frac{1}{2}F''a_1^2 + E'a_1 = 0,$$

which gives Equation (5.7). At order ε^3 ,

$$F'a_3 + F''a_1a_2 + \frac{1}{6}F'''a_1^3 + E'a_2 + \frac{1}{2}E''a_1^2 = 0,$$

and substitution gives Equation (5.8).

The general recurrence is Equation (5.11). To identify it with the Lagrange–Bürmann formula, note that both are triangular solutions with the same first coefficient. Formal uniqueness of the inverse in the ring of power series in ε forces equality. This direct argument is useful in a formal proof assistant because it reduces the theorem to coefficient extraction and uniqueness, without invoking contour integration.

D Implementation pseudocode

The following pseudocode separates the asymptotic seed, exact carrier refinement, and Bell-sector correction.

```

INPUT: large Y, carrier order K, sector order N, branch omega

A := log(e*Y/sqrt(2*pi))
w := LambertW_0(A/e)
Q := A/w
s := 1+w
u := Q - 1/2
u := u + d1(s)/Q + d3(s)/Q^3 + ... + dK(s)/Q^K

repeat until desired precision:
    f := logGamma(u+1) - 1 - log(Y)
    fp := digamma(u+1)
    u := u - f/fp

C := exp(-1) * sum_{j>=0} 1/(j!*(u+j+1))
lambda := digamma(u+1)

Phi[1] := -C/lambda
for n from 2 to N:
    H := C^n/lambda
    for r from 1 to n-1:
        H := ( derivative(H,u) + (n*omega-r*lambda)*H )/lambda
    Phi[n] := (-1)^n * H/n!

x := u + sum_{n=1}^N exp(n*omega*u)*Phi[n]/Y^n
residual := Gamma(x+1)/e + exp(omega*x)*C(x) - Y
OUTPUT: x, residual, residual-based error certificate

```

The residual is evaluated at the final approximant x . For the parity inverse, set $\omega = 0$ and multiply the n -th sector by σ^n . For the symmetrized real inverse, replace $E(x)$ by $C(x) \cos(\pi x)$ and use the general formula Equation (5.5).

E Reference table of principal formulas

Object	Formula
Branch continuation	$\mathfrak{D}_\kappa(z) = \Gamma(z+1)/e + e^{(2\kappa+1)\pi iz} C(z).$
Bell correction	$C(z) = e^{-1} \int_0^1 t^z e^t dt = e^{-1} \sum_{j \geq 0} [j!(z+j+1)]^{-1}.$
Bell asymptotic	$C(x) \sim \sum_{m \geq 1} (-1)^{m-1} \mathcal{B}_m x^{-m},$ with remainder bounded by $\mathcal{B}_{M+1} x^{-M-1}.$
Gamma carrier	$A = \log(eY/\sqrt{2\pi}), w = W_0(A/e), Q = A/w, s = 1 + w.$
Envelope inverse	$u = Q - \frac{1}{2} + (24sQ)^{-1} - (14s^2 + 10s + 5)/(5760s^3Q^3) + \dots.$
Additive inversion	$G^{-1} = u + \sum_{n \geq 1} (-1)^n \mathcal{D}_u^{n-1} (E(u)^n / F'(u)) / n!.$
Branch inverse	$z_\kappa = u + \sum_{n \geq 1} Y^{-n} e^{n\omega_\kappa u} \Phi_n^{(\omega_\kappa)}(u).$
First branch sector	$-Y^{-1} e^{\omega_\kappa u} C(u) / \psi(u+1).$
Real interpolation	$\mathfrak{D}_{\text{sym}}(x) = F(x) + C(x) \cos(\pi x).$
Discrete inverse	$N(Y) = \left\lceil \mathfrak{D}_{\text{sym}}^{-1}(Y) \right\rceil$ eventually.
Touchard deformation	$C_a(x) \sim \sum_{m \geq 1} (-1)^{m-1} \mathcal{T}_m(a) x^{-m}.$

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